

Spring Data & JPA Homework

Q1. List all of the annotations you learned

Common annotations include @Entity, @Table, @Id, @GeneratedValue, @Column, @Repository, @Service, @RestController, @RequestMapping, @GetMapping, @PostMapping, @PutMapping, @DeleteMapping, @Autowired, and @Transactional.

Q2. Comment feature of the class project

The Comment feature consists of an entity, repository, service, and controller. It supports basic CRUD operations for comments associated with posts.

Q3. Call all APIs in PostController and CommentController

All APIs can be tested using Postman by sending GET, POST, PUT, and DELETE requests and verifying the returned responses.

Q4. What is JPA and what is Hibernate?

JPA is a Java specification that defines ORM standards. Hibernate is an ORM framework that implements JPA and provides the actual persistence behavior.

Q5. What is Hikari? What are the benefits of connection pool?

HikariCP is a high-performance JDBC connection pool. It improves application performance by reusing database connections instead of creating new ones.

Q6. What is @OneToMany, @ManyToOne, @ManyToMany?

These annotations define relationships between entities. @OneToMany represents one-to-many, @ManyToOne represents many-to-one, and @ManyToMany represents many-to-many relationships.

Q7. What is cascade and orphanRemoval?

CascadeType.ALL propagates persistence operations from parent to child entities. orphanRemoval=true automatically deletes child records when they are no longer referenced.

Q8. What is FetchType.LAZY and FetchType.EAGER?

LAZY loads related entities only when accessed, while EAGER loads them immediately. LAZY is generally preferred to avoid unnecessary database queries.

Q9. What is the rule of JPA naming convention?

JPA allows query methods to be generated automatically based on method names, such as findById or findByUsername.

Q10. JPA advanced methods

JPA provides advanced query capabilities through repository naming conventions and built-in query generation.

Q13. What is JPQL?

JPQL is a query language that operates on entity objects instead of database tables and is independent of the underlying database.

Q14. What is @NamedQuery and @NamedQueries?

@NamedQuery defines a reusable JPQL query, while @NamedQueries allows multiple named queries to be declared on an entity.

Q15. What is @Query?

@Query allows developers to define custom JPQL or SQL queries directly inside repository interfaces.

Q16. What is HQL and Criteria Queries?

HQL is Hibernate's object-oriented query language. Criteria Queries provide a type-safe, programmatic way to build queries.

Q17. What is EntityManager?

EntityManager manages the lifecycle of entities and handles persistence operations.

Q18. What is SessionFactory and Session?

SessionFactory creates Session objects. Session represents a single unit of interaction with the database.

Q19. What is Transaction?

A transaction ensures data consistency by committing or rolling back operations as a single unit.

Q20. What is Hibernate caching?

Hibernate caching improves performance by reducing repeated database access through in-memory storage.

Q21. Difference between first-level and second-level cache

First-level cache is session-scoped and always enabled. Second-level cache is shared across sessions and must be explicitly configured.

Q22. How do you understand @Transactional?

@Transactional manages transaction boundaries automatically and ensures rollback on failure.